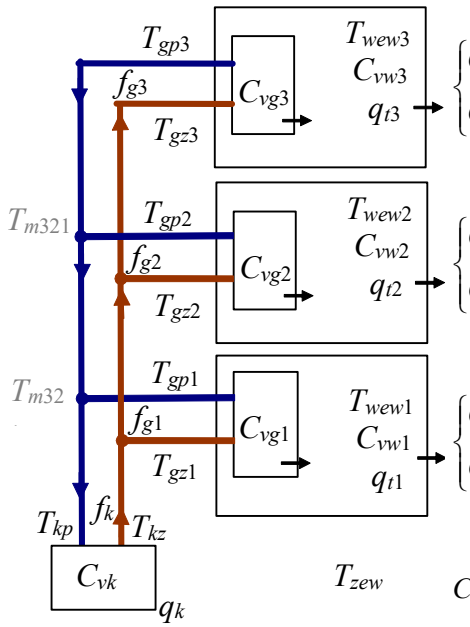
Budynek z trzema pomieszczeniami, z lokalnym źródłem ciepła (wersja KociolBud3CvwCvg)



$$\begin{cases} C_{vw3}\dot{T}_{wew3} = K_{cg3}\left(T_{gp3} - T_{wew3}\right) - K_{cw3}\left(T_{wew3} - T_{zew}\right) + q_{d3} \\ C_{vg3}\dot{T}_{gp3} = c_{pw} f_{mg3}\left(T_{kz} - T_{gp3}\right) - K_{cg3}\left(T_{gp3} - T_{wew3}\right) \end{cases}$$

$$\begin{cases} C_{vw2}\dot{T}_{wew2} = K_{cg2}\left(T_{gp2} - T_{wew2}\right) - K_{cw2}\left(T_{wew2} - T_{zew}\right) + q_{d2} \\ C_{vg2}\dot{T}_{gp2} = c_{pw} f_{mg2}\left(T_{kz} - T_{gp2}\right) - K_{cg2}\left(T_{gp2} - T_{wew2}\right) \end{cases}$$

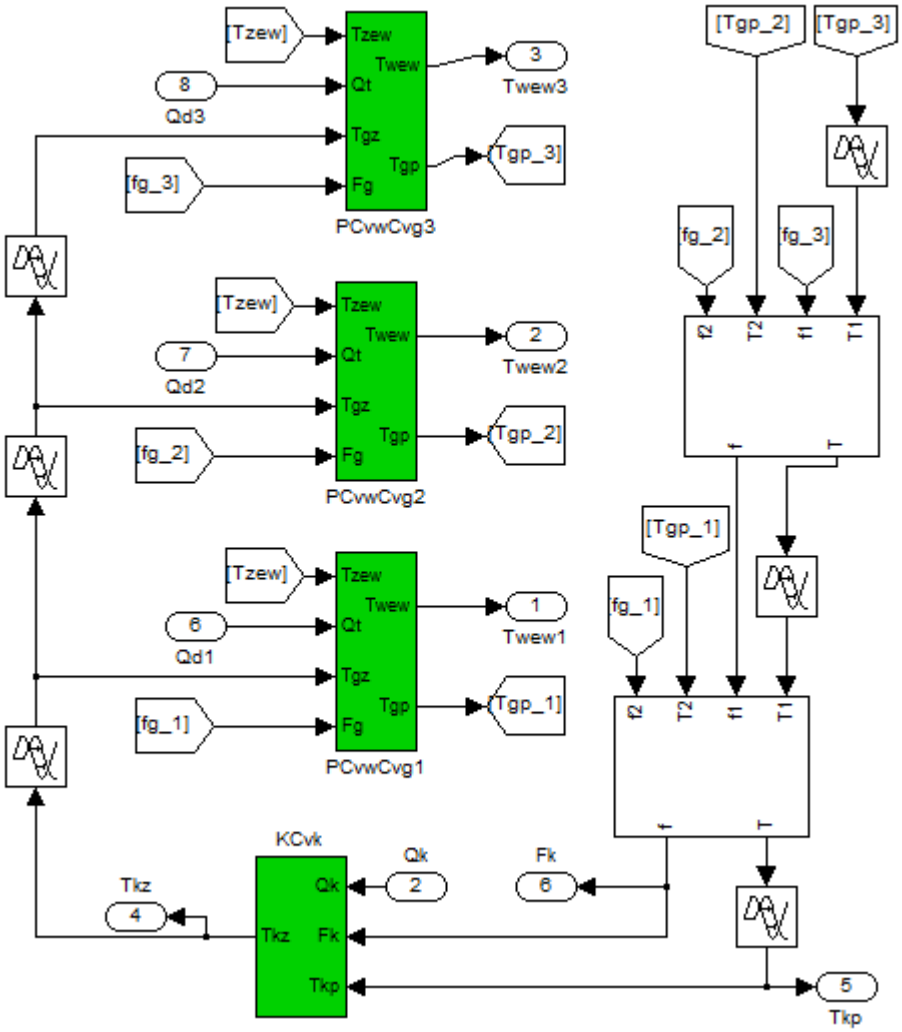
$$\begin{cases} C_{vw1}\dot{T}_{wew1} = K_{cg1}\left(T_{gp1} - T_{wew1}\right) - K_{cw1}\left(T_{wew1} - T_{zew}\right) + q_{d1} \\ C_{vg1}\dot{T}_{gp1} = c_{pw} f_{mg1}\left(T_{kz} - T_{gp1}\right) - K_{cg1}\left(T_{gp1} - T_{wew1}\right) \end{cases}$$

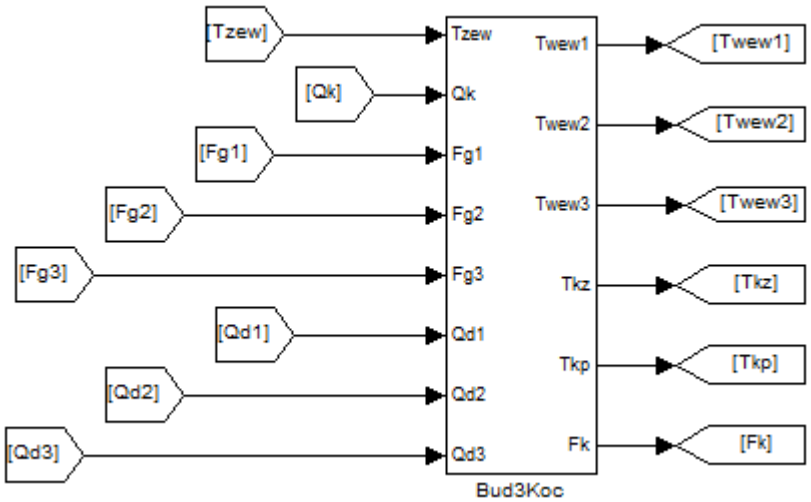
$$C_{vk}\dot{T}_{kz} = q_k - c_{pw}\left(f_{mg1} + f_{mg2} + f_{mg3}\right)T_{kz} + c_{pw}\left(f_{mg1}T_{gp1} + f_{mg2}T_{gp2} + f_{mg3}T_{gp3}\right.$$

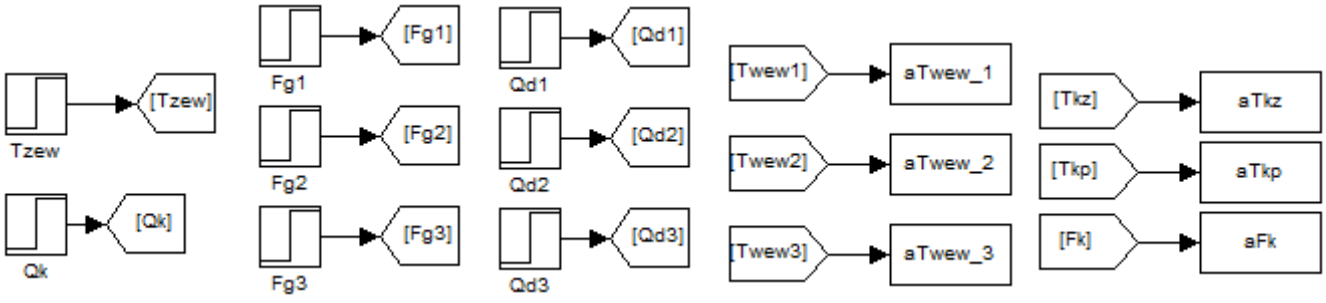
Razem

$$\begin{cases} C_{vw1}\dot{T}_{wew1} = K_{cg1}\left(T_{gp1} - T_{wew1}\right) - K_{cw1}\left(T_{wew1} - T_{zew}\right) + q_{d1} \\ C_{vg1}\dot{T}_{gp1} = c_{pw} f_{mg1}\left(T_{kz} - T_{gp1}\right) - K_{cg1}\left(T_{gp1} - T_{wew1}\right) \\ C_{vw2}\dot{T}_{wew2} = K_{cg2}\left(T_{gp2} - T_{wew2}\right) - K_{cw2}\left(T_{wew2} - T_{zew}\right) + q_{d2} \\ C_{vg2}\dot{T}_{gp2} = c_{pw} f_{mg2}\left(T_{kz} - T_{gp2}\right) - K_{cg2}\left(T_{gp2} - T_{wew2}\right) \\ C_{vw3}\dot{T}_{wew3} = K_{cg3}\left(T_{gp3} - T_{wew3}\right) - K_{cw3}\left(T_{wew3} - T_{zew}\right) + q_{d3} \\ C_{vg3}\dot{T}_{gp3} = c_{pw} f_{mg3}\left(T_{kz} - T_{gp3}\right) - K_{cg3}\left(T_{gp3} - T_{wew3}\right) \\ C_{vk}\dot{T}_{kz} = q_k - c_{pw}\left(f_{mg1} + f_{mg2} + f_{mg3}\right)T_{kz} + c_{pw}\left(f_{mg1}T_{gp1} + f_{mg2}T_{gp2} + f_{mg3}T_{gp3}\right) \end{cases}$$

$$\begin{cases} C_{vw1}\dot{T}_{wew1} = -\left(K_{cg1} + K_{cw1}\right)T_{wew1} + K_{cg1}T_{gp1} + K_{cw1}T_{zew} + q_{d1} \\ C_{vg1}\dot{T}_{gp1} = K_{cg1}T_{wew1} - \left(K_{cg1} + c_{pw} f_{mg1}\right)T_{gp1} + c_{pw} f_{mg1}T_{kz} \\ C_{vw2}\dot{T}_{wew2} = -\left(K_{cg2} + K_{cw2}\right)T_{wew2} + K_{cg2}T_{gp2} + K_{cw2}T_{zew} + q_{d2} \\ C_{vg2}\dot{T}_{gp2} = K_{cg2}T_{wew2} - \left(K_{cg2} + c_{pw} f_{mg2}\right)T_{gp2} + c_{pw} f_{mg2}T_{kz} \\ C_{vw3}\dot{T}_{wew3} = -\left(K_{cg3} + K_{cw3}\right)T_{wew3} + K_{cg3}T_{gp3} + K_{cw3}T_{zew} + q_{d3} \\ C_{vg3}\dot{T}_{gp3} = K_{cg3}T_{wew3} - \left(K_{cg3} + c_{pw} f_{mg3}\right)T_{gp3} + c_{pw} f_{mg3}T_{kz} \\ C_{vk}\dot{T}_{kz} = c_{pw}\left(f_{mg1}T_{gp1} + f_{mg2}T_{gp2} + f_{mg3}T_{gp3}\right) - c_{pw}\left(f_{mg1} + f_{mg2} + f_{mg3}\right)T_{kz} + q_k \end{cases}$$


[Tzew]
Tzew
Twew
Qt
Tgz
Tgp
Fg
PCvwCvg3
8
Qd3
3
Twew3
[Tgp_3]
[fg_3]
[Tzew]
PCvwCvg2
7
Qd2
2
Twew2
[Tgp_2]
[fg_2]
[Tzew]
PCvwCvg1
6
Qd1
1
Twew1
[Tgp_1]
[fg_1]
[Tgp_2]
[Tgp_3]
f2
T2
f1
T1
f
T
[Tgp_1]
[fg_1]
KCvk
Qk
Fk
Tkz
Tkp
2
6
4
5
Tkp


[Tzew]
[Qk]
[Fg1]
[Fg2]
[Fg3]
[Qd1]
[Qd2]
[Qd3]
Tzew
Qk
Fg1
Fg2
Fg3
Qd1
Qd2
Qd3
Twew1
Twew2
Twew3
Tkz
Tkp
Fk
[Twew1]
[Twew2]
[Twew3]
[Tkz]
[Tkp]
[Fk]
Bud3Koc


Tzew
[Tzew]
Qk
[Qk]
Fg1
[Fg1]
Fg2
[Fg2]
Fg3
[Fg3]
Qd1
[Qd1]
Qd2
[Qd2]
Qd3
[Qd3]
[Twew1]
[Twew2]
[Twew3]
aTwew_1
aTwew_2
aTwew_3
[Tkz]
[Tkp]
[Fk]
aTkz
aTkp
aFk

Badania podstawowe ((reakcje na wybrane zakłócenia): skrypt KociolBud_Badania

```
clear all; close all;
skala = 60;            %skala czasu (wszystkie czasy w minutach)
%gdy n=60, to Max step size dać 0.1
schemat = 'KociolBud3CvwCvg';
IlePom = 3;                  %ile pomieszczen
czas = 3000; czas_skok = 500;
%-------------------------------------------------------
kolor = 'rgbcm'; linia = ['- '; '--'; ': '];
KociolBud_skryptWN;                          %obliczenia parametrow (warunki nominalne), nazwa schematu
%model (wartosci parametrow z identyfikacji)
T = 1000; T0 = 100; k = 4 / (0.1*QkN);      %4 / 3000 = 0.0013
%=============================================================
%          BADANIA
%=============================================================
%============= warunki poczatkowe nominalne  ====================
Tzew0 = TzewN+1;
Qk0 = QkN*1;
Fg0 = [1, 1, 1] .* FgN;
Qd0 = [0, 0, 0];                             %zawsze
KociolBud_skryptP0;
%-------------------------------- zaklocenie --------------------------------
dTzew = 1;
dQk = 0*QkN;
dFg = [0, 0, 0] .* FgN;     %procent przeplywu nominalnego danego grzejnika
dQd = [0, 0, 0] .* QgN;      %procent zapotrzebowania nominalnego danego pomieszczenia
%-------------------------------------------------------
%symulacja                                          % wersja po 2019
[t] = sim(schemat, czas);                             %[out] = sim(schemat, czas);
figure(1); hold on;grid on;
plot(t, aTwew_1, strcat(kolor(1), linia(1,:)));     %plot(out,tout, out.aTwew_1, strcat(kolor(1), linia(1,:)));
plot(t, aTwew_2, strcat(kolor(2), linia(2,:)));
plot(t, aTwew_3, strcat(kolor(3), linia(3,:)));
%plot(t,aPV,'k');
title('T_w_e_w [^oC]'); xlabel('t[min]'); ylabel('Twew');
legend('T_w_e_w_1', 'T_w_e_w_2', 'T_w_e_w_3');

figure(2); hold on;grid on;
plot(t,aTkp,'b');
%plot(t,aTkp,'b','LineWidth',2);
title('T_k_p [^oC]'); xlabel('t[min]'); ylabel('Tkp');
```

Skrypt KociolBud_skryptWN (identyfikacja parametrów)

```
%=============wartosci nominalne globalne=====================
cpw = 4200; row = 1000;     %woda cp [J/kg K], ro [kg/m3]
cpp = 1000; rop = 1.2;          %powietrze cp [J/kg K], ro [kg/m3]
cps = 800; ros = 840;           %beton komorkowy cp [J/kg K], ro [kg/m3]
%---------- zapotrzebowanie i kociol --------------
QkN = 15*1000;           %moc kotla. 15 kW
Vk = 10/1000;         %pojemnosc kotla
Cvk = cpw*row*Vk;    %pojemnosc cieplna kotla
%----- wymiary budynku -------------------------------
VgN = IlePom * 50/1000;      %l -> m^3 (wszystkie grzejniki)
VwN = 10 * 10 * 2.5;             %m3 (caly budynek)
gs = 0.10;                           %grubosc scian
VsN = (10 + 2 * gs) * (10 + 2 * gs) * 2.5 - VwN;   %m3 (cały budynek)
 tau_siec = 240 / skala;          %opoznienia w sieci:  180 duze
 %-----------------------------------------------------
CvwN = cpp*rop*VwN + cps*ros*VsN ; CvgN = cpw*row*VgN; CvsN = 0;
CvwN = CvwN / skala; CvgN = CvgN / skala ; CvsN = CvsN / skala;  %przeskalowanie
 %---------------------------------------
TzewN = -20;
TwewN = 20; TsN = 5;
TgpN = 70;
TgzN = 90;
QdN = 0;
%---------- pomieszczenia (takie same) --------------
for i = 1:IlePom
    QgN(i) = QkN / IlePom;
    Kcg(i) = QgN(i) / (TgpN-TwewN);
    Kcw(i) = QgN(i) / (TwewN-TzewN);
    Cvg(i) = CvgN  / IlePom;     % grzejnik, zalezy od jednostki czasu
    Cvw(i) = CvwN / IlePom;     % pomieszczenie
    Cvs(i) = CvsN / IlePom;     % sciany
    FgN(i) = QgN(i) / (cpw*(TgzN-TgpN));
end
```

Skrypt KociolBud_skryptP0 (warunki początkowe)

```
%Na zewnatrz ustawione
%Tzew0, Qk0; Fg0 ; Qd0 = [0, 0, 0];
%------ obliczanie zmiennych stanu ------------------
s_123 = Fg0(1) + Fg0(2) + Fg0(3);
s_23 = Fg0(2) + Fg0(3);
s_12 = Fg0(1) + Fg0(2);
A=[-Kcg(1)-Kcw(1),  Kcg(1),                    0,0,                                    0,0,                                    0;
      Kcg(1),            -Kcg(1)-cpw*Fg0(1) ,   0,0,                                    0,0,                                    cpw*Fg0(1);
      0,0,                                      -Kcg(2)-Kcw(2),  Kcg(2),                0,0,                                    0;
      0,0,                                       Kcg(2),            -Kcg(2)-cpw*Fg0(2),  0,0,                                    cpw*Fg0(2);
      0,0,                                       0,0,                                    -Kcg(3)-Kcw(3),  Kcg(3),                0;
      0,0,                                       0,0,                                     Kcg(3),            -Kcg(3)-cpw*Fg0(3),  cpw*Fg0(3);
      0,cpw*Fg0(1),                              0,cpw*Fg0(2),                           0,cpw*Fg0(3),                           -cpw*s_123];
B=[Kcw(1), 0;
      0,         0;
      Kcw(2), 0;
      0,          0;
      Kcw(3), 0;
      0,0;
      0,         1];
u0 = [Tzew0; Qk0];
x0 = -A^-1*B*u0;
for i = 1:IlePom
    Twew0(i) = x0( (i-1)*2+1);
    Tgp0(i) = x0( (i-1)*2+2);
end
Tkz0 = x0( IlePom*2+1);
Tgp032 = (Tgp0(3)*Fg0(3) + Tgp0(2)*Fg0(2)  )  /  (  Fg0(3)+Fg0(2)    );              %TransportDelay
Tgp0321 = (Tgp0(3)*Fg0(3) + Tgp0(2)*Fg0(2)  + Tgp0(1)*Fg0(1)  )  /  (  Fg0(3) + Fg0(2) + Fg0(1)    ); %TransportDelay
Tkp0 = Tgp0321;
```

Badanie w różnych punktach pracy: skrypt KociolBud_BadaniaPunkty

```
clear all; close all;
skala = 60;            %skala czasu (wszystkie czasy w minutach)
%gdy n=60, to Max step size dać 0.1
schemat = 'KociolBud3CvwCvg';
IlePom = 3;        %ile pomieszczen
WariantModelu = 2;        %1= Cvw, 2= CvwCvg, 3= CvwCvgCvs,
czas = 3000; czas_skok = 500;
%-------------------------------------------------
kolor = 'rgbcm'; linia = ['- '; '--'; ': '];
KociolBud_skryptWN;                          %obliczenia parametrow (warunki nominalne), nazwa schematu (SKRYPT)
%model (wartosci parametrow z identyfikacji)
T = 1000; T0 = 100; k = 4 / (0.1*QkN);      %4 / 3000 = 0.0013
%=========================================================
%           BADANIA
%=========================================================
%============= warunki poczatkowe nominalne  ===================
%Punkty pracy:
Qd0 = [0, 0, 0];   %zawsze
tabTzew0 = [TzewN, TzewN+10, TzewN+20];
%podobnie tabQk0 ;tabFg10 ;tabFg20; tabFg30
Qk0 = QkN;
Fg0 = FgN;
%------------------ zaklocenie --------------------
   dTzew = 1;
   dQk = 0*QkN;
   dFg = [0, 0, 0] .* FgN;     %procent przeplywu nominalnego danego grzejnika
   dQd = [0, 0, 0] .* QgN;      %procent zapotrzebowania nominalnego danego pomieszczenia
%-------------------------------------------------------
for ipp = 1:3
   Tzew0 = tabTzew0(ipp);
    KociolBud_skryptP0;   %warunki początkowe (SKRYPT)
%symulacja                                          % wersja po 2019
   [t] = sim(schemat, czas);                          %[out] = sim(schemat, czas);
   figure(1); hold on;grid on;
   plot(t, aTwew_1, strcat(kolor(ipp), linia(1,:)));     %plot(out,tout, out.aTwew_1, strcat(kolor(1), linia(1,:)));
   plot(t, aTwew_2, strcat(kolor(ipp), linia(2,:)));
   plot(t, aTwew_3, strcat(kolor(ipp), linia(3,:)));
   %plot(t,aPV,'k');
   title('T_w_e_w [^oC]'); xlabel('t[min]'); ylabel('Twew');
   legend('T_w_e_w_1', 'T_w_e_w_2', 'T_w_e_w_3');

   figure(2); hold on;grid on;
   plot(t, aTwew_1 - Twew0(1), strcat(kolor(ipp), linia(1,:)));     %plot(out,tout, out.aTwew_1, strcat(kolor(1), linia(1,:)));
   plot(t, aTwew_2 - Twew0(2), strcat(kolor(ipp), linia(2,:)));
   plot(t, aTwew_3 - Twew0(3), strcat(kolor(ipp), linia(3,:)));
   %plot(t,aPV,'k');
   title('dT_w_e_w [^oC]'); xlabel('t[min]'); ylabel('dTwew');
   legend('dT_w_e_w_1', 'dT_w_e_w_2', 'dT_w_e_w_3');

end
```